A Flat Coherence Score: What a Production-Data Replay Does and Does Not Show About Agent Self-State Gating

Notes toward digital proprioception

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Abstract

A running multi-agent system I operate computes a scalar "coherence" for every agent check-in and uses it, among other inputs, to place each state in one of three basins (high, boundary, low). On a 30-day production slice of 13,292 rows, the deployed score was flat within a narrow band: its 1st to 99th percentile range was 0.447 to 0.499. It never reached the 0.40 threshold below which it could push a state into the low basin, and it cleared the 0.45 high-basin threshold for 98.7% of rows, so it almost never decided a basin. The score is not empty. It carries some class information (its distributions for two classes separate with an AUC of 0.88), and it correlates r = 0.73 with a smoothed energy-integrity imbalance. But within each class, its association with distance from that class's own healthy operating point was weak and went both ways: Spearman ρ from -0.37 to +0.18 across classes, with the wrong sign in two of five, and in three of five in a later window.

I replaced the score with a class-conditional "grounded" form on the same rows and recomputed every basin under both forms. The grounded form spans 0 to 0.995, and on the full substitution every row whose recomputed basin changes moves downward (§4). The replay is an offline counterfactual on frozen data. It shows that the old score was inert at its decision thresholds. It does not show that the new labels are right, and no outcome, harm or operator judgment was used to grade either form. The rate of label change rose from 28.8% to 44.3% between two public windows whose end dates are four to five days apart. That rise is not noise. Rows whose stored values occur in both windows flip at 30.3%, and rows that appear only in the later window, inferred to be recorded after 2026-04-18, flip at 80.6%. The flip rate therefore depends on how far the state distribution has moved from the frozen class anchors; it is not a fixed property of the formula swap.

Three adjacent results bound the claim (§6). Whether per-agent state forecasts bad outcomes is unresolved, pending one pre-registered read. A genesis-anchored lineage check meant to catch slow drift does not discriminate as instrumented. The cumulative-deviation integral that motivated earlier versions of this paper is specified but not computed by any production code path. A single-agent case study tests an apparent "delayed shut-down" reading: the April anchor's calibration window ended a day after a shift in Lumen's recorded state, and recalibration moves toward the current state on two of three coordinates, as staleness predicts. That result does not by itself exclude a persistent shift, and the cause is open (§5).

Keywords: agent state monitoring, coherence score, class-conditional calibration, counterfactual replay, calibration staleness, digital proprioception

1. Introduction

1.1 What this paper is about

The system is UNITARES (Wang 2026a). Each agent in a small production fleet checks in periodically, and the system estimates a four-coordinate state for it, called E, I, S and V. From that state it computes a coherence value, combines coherence with the other coordinates and a risk score, and classifies the state into a basin. The intent behind the design is what I have called digital proprioception: an agent-side analogue of the sense an organism has of where its own limbs are (Proske and Gandevia 2012), supplied to agents that otherwise have no such sense. The term imports a function, not a failure mode and not a claim about experience.

The question here is narrow. Did the deployed coherence score do the job it was given, which is to indicate how far an agent is from its own healthy operating point, at the thresholds the classifier applies to it? For the window I could measure, it did not. Everything else in the paper is a consequence of that fact, a test of what replacing the score does, or a statement of what I cannot yet claim.

1.2 Contributions

1. **A measured negative about the deployed score (§3).** The legacy coherence score had an observed range of 0.068 on 13,292 rows. It could not reach the low-basin threshold, it almost always cleared the high-basin threshold, and within each class it did not consistently track distance from the class healthy point, although its distributions do differ between classes.
2. **A reproducible replay with an exact window decomposition (§4).** Replacing the score changes recomputed basin labels for 28.8% of rows, always downward on the full substitution. The higher rate in a later window decomposes, by counting matched stored values, into rows that appear only in the later window (inferred to be recorded after 2026-04-18), which flip at 74% to 100% in four of five classes. The export and scripts are public.
3. **A calibration case (§5).** A single-agent case that looked like a failure mode from the allostatic-load literature is consistent with a calibration anchor whose window closed a day after a shift in recorded state. Recalibration is consistent with that reading but cannot exclude a persistent shift, and the cause remains open.
4. **A status ledger (§6).** Outcome forecasting is unresolved, not negative. The lineage check does not discriminate. The cumulative-deviation integral is specified and uncomputed.

1.3 What I do not claim

I do not claim that the replacement score is better than the legacy score (§4.4), that agents are conscious, that the system implements the biology of allostatic load, or that the cumulative-deviation integral has been tested. I do not claim that these results transfer to another deployment: the data come from one operator's fleet, a good part of which is a single embodied agent on one device. I do not claim the replay's flip rate is a stable property of the fleet.

2. The system and its instruments

2.1 State, coherence and basins

Each check-in writes a state vector (E, I, S, V) to a database relation. E is an energy or resource-rate proxy, I is an integrity proxy bounded in [0, 1], and S is an entropy or uncertainty proxy bounded in [0, 1]. V is the coordinate that needs care.

V is not an independent axis. The v6 specification describes V as an accumulator driven by the imbalance between E and I (Wang 2026a, Appendix A). Since April 2026 the production system surfaces a behavioral V instead: an exponentially smoothed E minus I. On the released rows of the first window, regressing V on the instantaneous E minus I gives r = 0.953 and $R^2 = 0.908$, with sd(V) = 0.081 against sd(V - (E - I)) = 0.026. The EISV vector is therefore better described as three state coordinates plus a smoothed imbalance term than as four independent dimensions. The older accumulator-style coordinate was demoted to a separate diagnostic, and that diagnostic still feeds the legacy coherence. Two different variables are therefore both called V in the codebase and in earlier drafts. I write V for the surfaced behavioral coordinate and say so when I mean the other.

The legacy coherence score. The fleet-wide form is

$$C_{\text{legacy}} = 0.5\,(1 + \tanh(V_{\text{void}}/V_{\text{scale}}))$$

with one fleet-wide $V_{\text{scale}} = 1$, where V_{void} is the demoted accumulator-style coordinate. (The code writes the scale as a gain, $C_1 = 1.0$, and multiplies by a ceiling $C_{\max} = 1$; both values are the same in the source on 2026-03-11 and on 2026-04-24, either side of the first window.) It reads that one coordinate only. The replay uses the stored legacy score and does not recompute it.

The grounded coherence score. The replacement reads (E, I, S) and measures distance to a class-specific healthy point μ_c, scaled by a class-specific envelope $\Delta_{\max,c}$:

$$C_{\text{grounded}}(\mathbf{x}, c) = \max\left(0,\ 1 - \|\mathbf{x}_{EIS} - \mu_c\|_2/\Delta_{\max,c}\right)$$

The class constants come from a 30-day healthy slice frozen on 2026-04-18: rows whose recorded regime label was nominal, STABLE, CONVERGENCE or EXPLORATION, labels the system under study itself produced. The healthy point is the coordinate-wise median of (E, I, S), and the envelope is the 95th percentile of distance from it (CIRWEL 2026, unitares, `scripts/calibrate_class_conditional.py`, commit c9cc38c). The two forms read different coordinates, so substituting one for the other replaces the signal, not a parameter of it. The two signals are not unrelated, though: the legacy score correlates r = 0.73 with the surfaced V, which is itself mostly E minus I.

Basin classification. The production `classify_basin` function returns high, boundary or low. Low is disjunctive: I < 0.5, C < 0.40, |V| > 0.30, or risk ≥ 0.70 each enter it. High is conjunctive: E ≥ 0.6, I ≥ 0.7, S ≤ 0.25, |V| ≤ 0.15, C ≥ 0.45 and risk ≤ 0.45 must all hold.

Boundary is the remainder. A change in coherence alone can move a row only when the other coordinates do not already force low or block high. In the current decision code, a low basin yields a `pause` and a boundary basin a `guide`, after higher-priority checks.

2.2 Classes

Class assignment checks the agent's label first: a named resident is its own class. Otherwise a tag decides (embodied, ephemeral, or persistent and autonomous), and an agent with none of these is in the default class, which the code describes as session-bounded agents and anything unrecognized. Five classes received frozen constants: Lumen (an embodied agent on a Raspberry Pi with environmental sensors), Sentinel, Vigil, Watcher (autonomous services), and default. A class without frozen constants falls through to a fleet fallback (the corner of the high basin, E = 0.6, I = 0.7, S = 0, with an envelope of 1.8); in the original pull only 42 `ephemeral` rows did (§4.2). The class constants, from the Phase 2 calibration (Wang 2026a §11.5; the healthy counts are those recorded in commit c9cc38c):

Class	Healthy N	E_h	I_h	S_h	$\Delta_{\max}$
Lumen	7,320	0.745	0.800	0.168	0.119
default	2,033	0.726	0.793	0.236	0.202
Sentinel	1,870	0.751	0.798	0.193	0.170
Vigil	384	0.737	0.790	0.240	0.171
Watcher	283	0.748	0.769	0.248	0.395

The envelope $\Delta_{\max}$ ranges over a factor of about 3.3 across classes. The fleet-wide constant used before per-class measurement (1.8) was 4 to 15 times larger than any measured envelope. The same-row ablation of §4.3 re-measured these constants on its own pull and obtained slightly different healthy counts and envelopes (for example default 0.200, Watcher 0.398). The replay uses the frozen values above. Lumen supplies 59% of the rows in the export, and Sentinel, Vigil and Watcher are each a single service, so the class structure rests on a small number of distinct sources.

2.3 The data

Everything in §3 and the full-substitution replay of §4 can be recomputed from a de-identified export archived under Zenodo data DOI 10.5281/zenodo.19705151 and mirrored in this repository. The §4.3 ablation cannot, for the reason given there. The export holds one row per agent-state observation, pseudonymized to a class label, with E, I, S, V, risk, both coherence values and both recomputed basin labels. It carries no agent identifiers, session identifiers, prompts or free text. Two windows are included: the 30-day window ending 2026-04-18 (13,292 rows) and the 30-day window ending 2026-04-23 (16,879 rows, Phase 2 constants held frozen). Matching on stored values and counting with multiplicity, 12,177 value tuples occur in both: 91.6% of the first window and 72.1% of the second. The match is by value, not by row identity, because the export has no row identifiers.

I report the 13,292-row export, not the 13,310-row pull that v6 cited. They are two executions of a wall-clock-anchored rolling window, and they differ by more than arrivals: by class, the export has Lumen −1, Sentinel −6, Vigil −1, Watcher −9, default +41, and none of the original

42 `ephemeral` rows, so 59 rows fewer and 41 more. The ephemeral −42 and default +41 suggest those agents were reclassified between the runs; I have not verified that. The production relation no longer retains the window (it held 490 rows in the interval when counted on 2026-08-14, against the 13,310 originally returned, and there is no archive table; the query is in `REPRO.md`), so the export is the surviving row-level record.

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3. The result: the deployed coherence score was inert at its thresholds

3.1 The numbers

All figures are computed from `verdict_counterfactual_v6_submission.csv` (n = 13,292) by `flatness_and_windows.py`.

	min	p1	p50	p99	max	sd
C_legacy	0.441	0.447	0.481	0.499	0.508	0.0082
C_grounded	0.000	0.000	0.625	0.983	0.995	0.2748

The legacy score occupies a band 0.068 wide, with an interquartile range of 0.4799 to 0.4847. No row falls below 0.40, so the $C < 0.40$ clause of the low-basin rule was unreachable for this fleet in this window. 98.7% of rows satisfy the $C \geq 0.45$ clause of the high-basin rule, so that clause was close to a constant true. The coherence input therefore almost never decided a basin. Setting coherence to 1 for every row, so that both of its clauses always pass, changes 6 of the 13,292 recomputed basins (0.045%).

The grounded score spans the unit interval. 6.8% of rows sit exactly at zero, 15.5% are below 0.25, and 26.6% are below 0.40.

3.2 It did not track distance from the agent's own healthy point

The legacy score correlates r = 0.73 with the surfaced V, and it carries some class information: the probability that a random row of one class scores above a random row of another (the AUC) ranges from 0.54 to 0.88 across class pairs, highest for Lumen against default.

A working distance-from-healthy-point score should fall as a state moves away from its class's healthy point, within every class. The legacy score did not. Far rows have C_grounded < 0.1; near rows have C_grounded > 0.8:

Class	n	Spearman ρ, legacy vs distance	Legacy p50, far rows	Legacy p50, near rows
Lumen	7,889	-0.27	0.480 (n = 646)	0.482 (n = 2,343)
Sentinel	2,221	+0.18	0.472 (n = 161)	0.482 (n = 343)
Vigil	471	+0.07	0.495 (n = 37)	0.483 (n = 76)
Watcher	354	-0.27	0.456 (n = 72)	0.485 (n = 45)
default	2,357	-0.37	0.485 (n = 306)	0.485 (n = 169)

The association is weak and goes both ways: the wrong sign in two of five classes here, and in three in the second window (Lumen +0.24, Sentinel +0.50, Vigil +0.52). Where the sign is right, the size is not usable: the largest gap between far and near medians is 0.03 (Watcher), and the far rows still sit above the 0.40 low-basin threshold. A Lumen state at the edge of its envelope and one at its centre receive scores 0.002 apart.

3.3 What this does and does not establish

The result is a statement about the legacy instrument at its deployed thresholds: it could not reach one of them, it almost always cleared the other, and it did not consistently register distance from a class's healthy point.

It does not say the grounded score's spread is signal. Part of it is arithmetic: Lumen's envelope is 0.119, so a state displaced by 0.06 from the healthy point already scores 0.5. Dividing by a small radius widens the spread by construction.

It does not say why the legacy score is narrow. The median legacy score in this window corresponds to a tanh argument of -0.037, close enough to zero that tanh is near its midpoint. I have not isolated why the demoted coordinate sat there.

It does not generalize beyond this fleet. The second public window gives a legacy median of 0.483 and a 1st to 99th percentile of 0.450 to 0.500, the same picture, but it shares 12,177 matched rows with the first and is not independent.

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4. The replay

4.1 Design

For every row, I compute the basin twice with the same `classify_basin` function: once with the stored legacy coherence and once with the grounded coherence computed against the frozen Phase 2 class constants. Both basins are recomputed by the replay from stored columns of the state relation (E from the stored state record, and I, S, V and the stored coherence). They are not read from persisted production verdicts, and I have not checked that these stored columns are identical to the inputs the live decision path saw. A flip is a row whose recomputed basin differs. This is a static reclassification of fixed state vectors under two formulas. It is not a test of a deployed mechanism, and no agent's behavior changed.

4.2 Result

On the public export of the first window, 28.8% of rows (3,834 of 13,292, counted from recomputed labels; the original 13,310-row pull gave 28.9%) would change recomputed basin when each class is measured against its own healthy point instead of the fleet-wide formula. Per-class rates range from 15.8% (Sentinel) to 33.5% (Vigil).

- **The direction is downward.** In the full substitution, every flipping row moves toward the low basin: 1,702 boundary to low, 1,404 high to low and 725 high to boundary, counted from the labels stored in the export. No row moves up. This is mechanical. The legacy score cannot reach the $C < 0.40$ clause and the grounded score reaches it for more than a

quarter of rows, so the replay mostly records how often a score that can cross a threshold does so, against one that cannot.

- **The later window decomposes by composition.** The window ending 2026-04-23 gives 44.3%. Because recomputed labels are a deterministic function of the stored values and frozen constants, a value tuple present in both windows flips the same way in both, and the difference is entirely composition. Rows are matched on their stored values, counted with multiplicity: a multiset intersection, not a row-by-row join. The export has no row identifiers, and the match uses seven columns (class, E, I, S, V, risk and the legacy score; the grounded score is left out because it is computed from unrounded coordinates). On that key the first window has 377 groups of rows that share all seven values, holding 1,228 rows. Rows in a group carry identical stored labels in both windows, so the counts are unaffected, but a match does not establish that both windows contain the same observation. The export also has no timestamps, so the timing in the row labels below is inferred from window membership. The flip rates use the labels stored in the export:

Rows	n	Flip rate
In the first window only (inferred: its earliest rows)	1,115	13.1%
In both windows	12,177	30.3%
In the second window only (inferred: recorded after 2026-04-18)	4,702	80.6%
of which Lumen	1,694	100.0%
of which Sentinel	1,023	96.1%
of which Vigil	173	96.0%
of which default	1,288	73.6%
of which Watcher	524	0.0%

In three classes, second-window-only rows flip almost without exception, and most of them fall below the 0.40 grounded threshold (Lumen 100%, Vigil 87%, Sentinel 70%). Far fewer lie outside the frozen envelope altogether, at C_grounded = 0 (16.6%, 12.1% and 3.1%). The flip rate therefore tracks how far the current state distribution has moved from the frozen anchors. That is what anchor staleness would produce, but flips also depend on the other basin coordinates and on risk, so the rate is not a direct measure of staleness. A row-wise Wilson interval (about $\pm$0.8 points within one window) assumes independent observations. Repeated measurements of a few agents do not justify that assumption, and the interval does not describe between-window composition changes. The first-window rate is a property of that window and those anchors, not of the formula swap in general. If second-window-only rows were all recorded between the two window ends, they would imply a higher row rate: 4,702 rows in four to five days, against about 440 per day across the first window. Without timestamps or row identifiers, that rate change remains inferred.

- **Two measurement caveats from the original pull.** The 30-day window overlaps identity-system revisions in mid-to-late April 2026, so some rows may carry class assignments inherited from cached bindings or from archived predecessors. And 42 rows of an

`ephemeral` class fell through to the fleet fallback and are not interpreted; the export carries none of them (§2.3).

4.3 A same-row ablation, with a boundary

The replay changes two things at once: the formula (tanh of the void coordinate, versus distance in E, I, S) and the calibration target (one fleet-wide constant, versus per-class constants). I separated them on the same 13,310 rows from the original pull, where GF is the grounded formula against a single fleet-wide healthy point and radius measured from the same slice:

Comparison	Flips / N	Rate
Legacy → grounded fleet-wide (formula change)	1,496 / 13,310	11.2%
Grounded fleet-wide → grounded class-conditional (calibration change)	3,133 / 13,310	23.5%
Legacy → grounded class-conditional (full substitution)	3,844 / 13,310	28.9%
Legacy → class-scaled tanh (artificial control)	10,355 / 13,310	77.8%

The terms do not add (11.2 plus 23.5 is not 28.9) because the effects interact. I read the grounded fleet-wide to class-conditional step as evidence that the class-envelope term has an effect on basin labels of its own, and I do not decompose the headline further. The fourth row grafts a per-class scale onto a tanh form, but applies it to the surfaced V rather than to the demoted coordinate the production score reads, so it changes the input as well as the scale. It is unstable and I treat it as a negative control, not as a calibration-only path. The GF and class-scaled conditions need a per-row `regime` column that the de-identified export does not carry, so those two rates are provenance-backed from the recorded ablation output (`formula_calibration_ablation_results.txt`) and are not independently re-runnable.

The calibration-change step is not uniformly downward: 373 rows in it move up (365 low to boundary, 5 low to high, 3 boundary to high). The all-downward statement applies to the full substitution only.

4.4 What the replay shows

It shows that the formula and calibration choices matter at the basin level, and that the legacy score almost never decided a basin. It is recomputable by a third party from public rows and public scripts: `reproduce_basinflip.py` reproduces 26,574 of 26,584 basin labels exactly, with the ten exceptions within 6.7×10^{-4} of a threshold, the rounding floor of a four-decimal export.

It does not show that any flip is correct. Under the current decision code, a recomputed flip into the low basin corresponds to a `pause`, and a flip from high to boundary to a `guide`. Whether that is desirable depends on the agent and the context, and no outcome grades it. The replay is evidence about the old score and is silent about the new one.

4.5 Why the homogenization framing is only an analogy

The class-conditional step was motivated by the neuroscience literature on group-average versus individual-level measurement (Krakauer et al. 2017; Gratton et al. 2018): pooling subjects with

different dynamics yields a reference representative of none of them. The analogy names a failure mode. It is not a test. In the fleet the pooling was not the main defect. The main defect, per §3, is that the legacy score did not act at its thresholds, pooled or not. Class-conditional calibration is one remedy among several, and the replay does not rank it against the others. §4.2 also shows its cost: class anchors frozen at one date stop describing the fleet as it changes.

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5. A calibration case on one embodied agent

5.1 The apparent failure mode

McEwen's four types of allostatic load (McEwen 1998; 2007) describe four ways the normal stress response fails: repeated hits, lack of adaptation, delayed shut-down, and inadequate response with compensatory hyperactivity. On 2026-05-09 I observed Lumen across an 86-minute window of 27 check-ins. Against its April healthy point, the state looked like the delayed shut-down pattern. The behavioral V sat at 0.0954 with a within-window standard deviation of 0.00015, sign-flipped from the healthy value of about -0.055. The (E, I, S) displacement was 0.115 against an envelope of 0.119, so the grounded coherence was about 0.03, while the legacy coherence read 0.497. That contrast is itself an instance of §3: the legacy score called a state at 97% of its class envelope essentially neutral.

5.2 The disambiguation and its result

The case study specified a test in advance: if the healthy point moves toward Lumen's current state on recalibration, read the elevation as calibration staleness, not a persistent failure. I recalibrated on the 30-day window ending 2026-05-09 (11,472 rows). The record written that day describes the method as means over sessions with no pause or reject verdict, which is not what the Phase 2 script does (§2.1), so the two calibrations may not be comparable. Against the observed state in §5.1 (E 0.791, I 0.695, S 0.159), the reported healthy point moved toward it on two coordinates (E from 0.745 to 0.7706, I from 0.800 to 0.7335) and away from it on the third (S from 0.168 to 0.1893), and the envelope grew from 0.119 to 0.1279. On two coordinates of three, that is the result the criterion treats as staleness. The criterion cannot carry the conclusion, though: a reference refit on data from a persistently shifted state would also move toward that state, so the test separates the two readings only if the newer window is itself healthy, and nothing independent of the system confirms that. It would also pass if the shift were in the instrument rather than the agent, which §5.3 shows is possible.

Weekly bins of Lumen's state over 90 days show three regimes, and sub-day resolution places the change at a single ten-hour event on 2026-04-17 UTC (E falling from 0.749 to 0.350 and recovering to 0.787), after which the new regime held for 22 days. About 96.7% of the Phase 2 window's duration (2026-03-19 to 2026-04-18) predates the event. The share of its rows is not recoverable, because the exports carry no timestamps. How much of the anchor's data comes from each regime is therefore unknown. What is known is that the new regime spans at most the window's last day. The second-window rows in §4.2 show that the anchor does not describe the later state: every second-window-only Lumen row flips and falls below the 0.40 grounded threshold against the April anchor, and 16.6% of them lie outside its envelope altogether.

5.3 Cause: open

Sixteen pull requests merged into the governance server on 2026-04-17 (CIRWEL 2026, unitares, #8 to #24 except #22, which was closed unmerged; merge times in UTC from GitHub). Lumen's E fell between 00:00 and 09:00 UTC and rose into the new regime between 11:00 and 14:00. Four merges changed identity handling, but none is shown to have touched Lumen. #8 at 02:25 severed fingerprint-based inheritance of EISV state, and its description reports a check that found no effect on Lumen, which resumes by an explicit identifier. #14 at 08:42 removed name-based identity resolution, #19 at 11:03 changed only another resident, and #16 at 11:32 concerned explicit cross-agent targeting. Two merges touched Lumen's side of the server. #15 at 11:33 removed a fallback that could route sensor-derived EISV into the state record, and deleted the task that synced it; its description calls the fallback redundant with Lumen's direct push, which it kept. #20 at 12:32 moved the server's Mac-to-Pi orchestration tools into a separate plugin, which is not shown to have changed how Lumen's check-ins arrived. These are merge times. Deploy times are not retained. #13's description reports restarting the governance service that day, so at least one restart occurred, at a time not recorded.

The explanations divide into a change in the agent and a change in the instrument, and the record cannot separate them. There are three instrument candidates. #15, merged at the start of the recovery, would have changed Lumen's readings only if some of its check-ins still went through the removed fallback, which is not known. A restart can produce a trough: an operational note of 2026-06-03, which is not public, records a server restart resetting Lumen's server-side trajectory to a fallback in which E sat near 0.30. Whether that mechanism existed in April is not verified; the trough on 2026-04-17 was 0.35. The third is a CPU artifact. The E coordinate derives in part from CPU utilization, and a day of deployments generates load. At the time, a single CPU reading entered two anima dimensions, suppressing E and inflating I at once, a defect repaired in August 2026 (CIRWEL 2026, anima-mcp, PRs #173 and #176). A deploy-driven CPU excursion amplified by that defect could produce an energy collapse and recovery on this timescale with no change in the agent. The identity changes are a weaker candidate than any of these, given #8's own check. The CPU and memory series for that day are no longer retained, so I cannot re-derive E with the corrected mapping.

One observation from the public export bears on this without settling it. Second-window-only rows flip at 96% for Sentinel and Vigil, as well as 100% for Lumen, and fall below the 0.40 grounded threshold at 70% (Sentinel) and 87% (Vigil), as well as 100% (Lumen) (§4.2). The CPU defect was in Lumen's sensor mapping. A shift shared by the autonomous services points toward a server-side change. #15's description also says it fixed a latent bug by which one agent's sensor values could be inherited by another, so it may bear on the other classes as well. The export has no timestamps, so this cannot place the other classes' shift on 2026-04-17.

Two further limits. The recalibration windows are not sampling-matched: the Phase 2 Lumen healthy slice held 7,320 rows and the recalibration window 11,472, and I have not confirmed that both counts apply the same healthy filter. The higher row rate inferred in §4.2 could account for part of the difference. The production server flagged no anomaly throughout, but the deployment's degradation paths fail toward healthy rather than toward unknown, so the silence carries little weight.

What survives every one of these explanations is narrower: the April anchor's window closed a day after the shift, and the anchor no longer describes Lumen's current readings. Whether the

new state is healthy is not established. The case is a single-agent report. Its date, magnitude and shape are provenance-backed, its causal reading is open, and the longitudinal data that would test it have aged out of the production database. It is not evidence for or against McEwen's taxonomy in biology. It does suggest a category the four types do not cover, a shift in the operating point itself, but one case does not establish a category, and if the shift was in the instrument it is not a property of the agent at all.

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6. Three adjacent questions: status

None of these has reproducible evidence in this repository, so each is stated as a status with a pointer to its source. The longer treatment is §§6 to 8 of this repository at commit 284dd21, which is v2.0 with later corrections.

Outcome forecasting: unresolved. Whether per-agent state predicts bad outcomes is unresolved, not negative. A historical weekly ablation that appeared to show a negative result was withdrawn for target inference on 2026-08-26: its cohort mixed anchor populations the confirmatory read excludes, and its permutation blocks were not independent adjudicated failures (CIRWEL 2026, unitares, `docs/ontology/eisv-proprioception-contract.md`, item 4). I report no forecasting effect size, direction or sensitivity bound. One confirmatory read is pre-registered for 2026-12-01 (CIRWEL 2026, unitares, PR #1425; the registered text is `docs/proposals/registered/eisv-outcome-grounding-stop-rule-v0.md`). It passes only if all four of its conditions hold, one of which is at least 150 bad clusters on the read's trusted slice (prior-state permutation blocks, which the registration says do not establish that outcomes are independent). The evidence ledger records interim access to discrimination results by automation after registration, so the read is not a single clean blinded look. Whatever it shows, every bad outcome available so far is task-negative, which bounds even a positive result to rework prediction, and a discrimination statistic is not decision value at the deployed operating point. §3 does not depend on any of this.

Trajectory identity: does not discriminate as instrumented. The companion framework (Wang 2026b) adds a fixed genesis signature to the rolling reference, to catch slow drift that step-to-step comparison passes, with a similarity threshold of 0.60. As deployed, a genesis signature can be reseeded until the agent reaches the second trust tier, so it is an early-data estimate that hardens with tenure, not a captured pre-illness baseline. No genesis was persisted for Lumen at first onboarding (Wang 2026a §11.7, item 5), so the §5 case could not use one. The similarity metric does move with drift, but it does not separate agents at its threshold. For agents whose stored signatures carry a covariance, the composite saturates near 0.633, above the 0.60 cut, so the check cannot fire for them (CIRWEL 2026, unitares, `docs/ontology/eisv-proprioception-contract.md`, trajectory-identity item 9). Across all identities with both signatures, the median similarity to genesis is 0.996 below 20 observations and 0.631 at 1,000 or more (CIRWEL 2026, unitares, `src/trajectory_identity.py`, commit e4239e8a0): identities decay into that floor as drift accumulates, rather than crossing the line. The companion paper's v0.15 correction (Wang 2026b, §6.5, tag `paper-v0.15`) reports between-agent similarity of 0.63 against 0.12 for one agent compared with itself across a client migration, and returns its discrimination criterion to open. The two-tier architecture stands as a proposal that its instrumentation does not realize.

The cumulative-deviation integral: specified, not computed. Earlier versions of this paper were titled around an integral in the spirit of allostatic load (McEwen and Stellar 1993): for the embodied agent, the accumulated distance of a four-component anima vector (warmth, clarity, stability, presence, computed from sensors and system metrics) from the attractor center of its own recent states. The anima vector is a different object from the EISV vector of §2, and the integral is a different accumulator from either V. A function that computes it exists in the embodied agent's repository as a research diagnostic, with unit tests and no caller outside them (checked against the main branch on 2026-10-02). No production path computes it, thresholds it or acts on it, and the coupling the companion framework specifies (when the integral exceeds a deployer-set multiple of the basin scale, induce rest, reduce stimulation or pause a task) is not wired. Earning it would take wiring the coupling and showing that it improves decisions over a non-integral baseline such as instantaneous deviation. §§4.2 and 5 add a prior problem: an integral against an anchor that no longer describes the agent would accumulate that gap.

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7. Limitations and what would change the picture

One fleet, one operator. The data come from a single deployment, and over half the rows in the replay are one embodied agent. Class-level differences rest on a small number of distinct sources. An independent re-measurement on another deployment would be the stronger bar.

Retention. The production relation no longer holds the measurement window. The frozen export is the record, the Lumen longitudinal pull is foreclosed, and the causal reading of §5 cannot be upgraded from this deployment's history. A deployment used as a test bed needs its evidentiary windows pinned as exports at measurement time.

Same-author, self-cited system. The author built the system, ran the measurement and wrote the paper. The public export and offline recomputation reduce the dependence on trust for the §3 and §4 numbers. They do not reduce it for the framing.

The replacement score is unvalidated. Showing that the old instrument did not act at its thresholds is a different result from showing the new one is right. Testing the replacement needs an outcome-graded comparison, with a label source wider than task-negative outcomes and a decision-value analysis, and the available data do not support one.

Anchor staleness. The replay's rate depends on how far the state distribution has moved from frozen anchors (§4.2). Any deployment of class-conditional calibration would need a recalibration policy, which this paper does not test.

What would change the picture. If the December 1 read passes all four registered conditions, it meets the operational continuation gate: outcome-grounding remains open and Stage B may be reconsidered. Report the observed discrimination with the specified cohort, dependence, power and interim-access disclosures. PASS does not itself establish decision benefit. If the read fails, report the failed conditions and honor closure for this measurement premise; insufficient support is not scientific disproof. Reopening requires a materially different label channel or measurement process under the registered rule, not more of the same labels. If an independent deployment shows a legacy-style score that reaches its thresholds and tracks within-class distance, the §3 result is specific to this system's demoted coordinate.

8. Conclusion

The deployed coherence score was flat within a band from 0.447 to 0.499 (1st to 99th percentile). It never reached the 0.40 low-basin threshold and almost always cleared the 0.45 high-basin threshold, so it almost never decided a basin. It carried some class information, but within each class it did not consistently track distance from that class's healthy point. Replacing it moves many recomputed basin labels downward in an offline replay, and the rate of that movement depends heavily on how far the fleet has moved from the frozen class anchors. That shows the old score was inert where it mattered.

The single-agent case is consistent with a calibration anchor whose window closed a day after a shift in recorded state, but neither a persistent shift in the agent nor an instrument change is excluded. Of the adjacent questions in §6, outcome forecasting is unresolved, the lineage check does not discriminate, and the integral is not computed in production.

The practical lesson is about instruments. A scalar that looks like a health score on a dashboard can sit where its thresholds never touch it, and the way to find out is to measure its spread against the thresholds and against the thing it is supposed to distinguish. The next step for a replacement is an outcome-graded comparison, a tested recalibration policy, and a label source suited to the intended claim before describing it as a validated health indicator or using claimed benefit to justify a policy change.

Appendix A: Reproducibility

Object	Status	What can be checked now	What is missing
Flatness and within-class statistics (§3)	**Recomputable offline**	Flatness script: percentiles, threshold shares, within-class Spearman, far/near medians, class AUCs, both windows	Independent deployment
Full-substitution replay (§4.2)	**Recomputable offline**	Replay script: 28.84% (3,834 / 13,292 from recomputed labels; the labels stored in the export give 3,831), per-class rates within 0.6 points of the original pull, 26,574 of 26,584 labels exact	Independent deployment

Object	Status	What can be checked now	What is missing
Window decomposition (§4.2)	**Recomputable offline**	Flatness script, matching stored values with multiplicity: a multiset count, not a row-level join (no row identifiers exist)	Row identifiers; timestamps
Formula-versus-calibration ablation (§4.3)	Provenance-backed only	Recorded ablation output	Needs a per-row `regime` column absent from the export; the production window is no longer retained
Lumen recalibration case (§5)	Provenance-backed only	86-minute protocol, recalibration criterion, reported recalibration values and reported summaries of the weekly bins, as recorded in this repository on the day of measurement (commits ebbe7d3 and f03cfc6, 2026-05-09); merge times of the 2026-04-17 pull requests (GitHub)	The raw samples and the recalibration query output were not kept; state history for 2026-02 to 2026-04 aged out; CPU and memory series, deploy times and restart times for 2026-04-17 not retained
Cumulative-deviation integral (§6)	Specified; function exists as an uncalled research diagnostic	Integral function and its tests in the embodied agent's repository	Any production use
Outcome forecasting (§6)	Unresolved	Ontology contract item 4 and the pre-registered stop rule	The 2026-12-01 read
Row-level export	**Public**, Zenodo data DOI 10.5281/zenodo.19705151, mirrored here, SHA-256 pinned in the replay script	13,292 and 16,879 class-pseudonymized rows with state, risk, both coherences, both recomputed basin labels	Nothing for §3 and §4.2

Object	Status	What can be checked now	What is missing
Raw production relation	Withheld and no longer retained	Schema and provenance	Unavailable in principle

The flatness script is `flatness_and_windows.py`; the replay script is `reproduce_basinflip.py`. Both are in `analysis/phase-2-2026-04-18/` and use the standard library only. Recorded ablation output is `formula_calibration_ablation_results.txt` in the same directory. The integral function is `compute_void_integral` in the embodied agent's repository.

Appendix B: Corrections

v2.0.1

- §7 now states the registered consequence of December PASS: outcome-grounding remains open and Stage B may be reconsidered. PASS does not automatically validate forecasting or establish decision benefit; FAIL closes the measurement premise without making insufficient support scientific disproof.
- §§4.2 and 5 consistently describe second-window-only rows by membership. Their dates and any implied row-rate increase remain inferred because the exports contain no timestamps or row identifiers. Appendix B limits the missing-genesis statement to first onboarding.
- §8 now distinguishes the retained descriptive information from validation as a health indicator or evidence of policy benefit. The replay results, thresholds and registered conditions are unchanged.

Corrections to v1.2

- v1.2 called outcome forecasting "measured and negative". That claim was withdrawn on 2026-08-26 (§6). The question is unresolved.
- v1.2 said in one place that the genesis signature is "captured trivially" and in another that Lumen's was not persisted. §6 states what is stored, when it can be reseeded, and that no genesis was persisted for Lumen at first onboarding.
- v1.2's title named a cumulative-deviation integral that is not computed in production (§6).
- v1.2 described the two public windows as roughly 87% overlapping in rows and read the change in flip rate between them as window sensitivity. The row overlap is 91.6% of the first window and 72.1% of the second, and the change is a deterministic composition effect (§4.2).
- v1.2 described the Phase 2 procedure as means over sessions with no pause or reject verdict. The calibration script selects rows by regime label and uses medians (§2.1). The May recalibration was recorded under the v1.2 description, so its comparability is uncertain (§5.2).
- v1.2 read the Lumen anchor as averaging across a regime change. About 96.7% of its window's duration predates the change, so the shift came in the window's last day; the share of the anchor's rows from each regime is not recoverable (§5.2).
- The clinical study sketches, the kintsugi section, the differentiation survey and the synthetic-psychology essay are cut. They remain in the v1.2 tag.

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